

- 7 (a) Distinguish between the appearance of emission and absorption line spectra.

Emission spectrum consists of coloured lines on a dark background while absorption spectrum consists of dark lines on a coloured spectrum.

[2]

- (b) The lowest six discrete energy levels for a hydrogen atom are shown in Fig. 7.1, where the ground state is -13.6 eV.

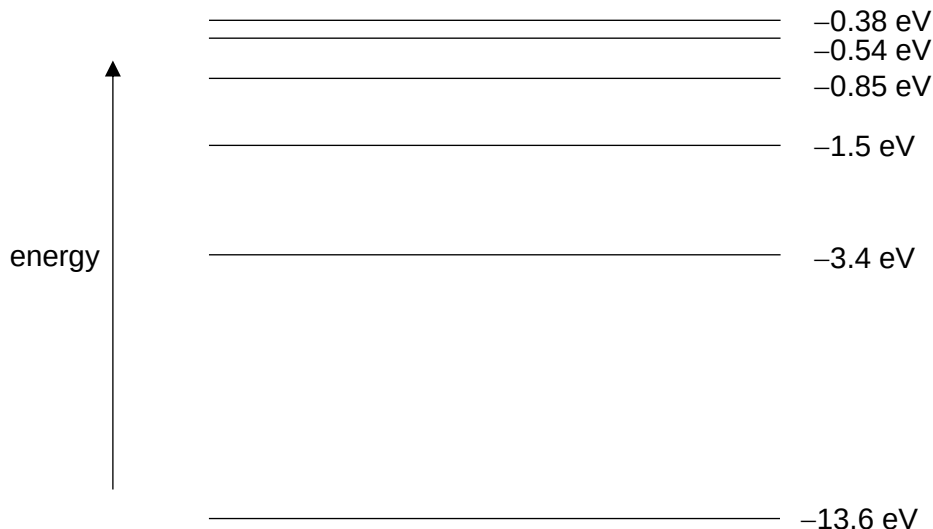


Fig. 7.1 (not to scale)

- (i) The spectrum produced by hydrogen is a line spectrum. Use Fig. 7.1 to explain why the spectrum is a line spectrum rather than a continuous spectrum.

Fig. 7.1 shows that the energy levels of electrons within the atom are discrete. Hence the energies of the photons emitted when the electrons de-excite from higher to lower energy levels are discrete as well. Light of a single wavelength and frequency is produced, corresponding to each line on the line spectrum.

[2]

- (ii) Describe one way by which an electron in gaseous hydrogen can be raised from a ground state to the -0.54 eV energy level.

Electrons can be excited through collision with other particles that possess at least 13.06 eV of energy. OR

Electrons can be excited through absorption of photons with exactly 13.06 eV of energy.

[1]

- (iii) State the total number of different wavelengths that may be emitted as the electron de-excites from the -0.54 eV energy level.

$$5C2 = 10$$

number = [1]

- (iv) Electromagnetic radiation is emitted when an electron falls to the ground state from the -0.54 eV energy level.

Calculate the wavelength of this radiation. Suggest the type of radiation emitted.

$$|E_f - E_i| = \frac{hc}{\lambda}$$

$$|(-13.6 + 0.54) \times 1.60 \times 10^{-19}| = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{\lambda}$$

$$\lambda = 9.52 \times 10^{-8} \text{ m}$$

ultraviolet

wavelength = m

type of radiation =

[2]

[Total: 8]

2 Read the passage below and answer the questions that follow.